Neutrino Mixing Angles from the Mesh Framework (PMNS)

Mixing-Angle Estimates and Their Status

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper reports the mesh framework's treatment of the neutrino mixing (PMNS) angles. The framework offers structural estimates of the mixing angles from the relative geometry of the neutrino knot phases. We report these as estimates with their comparison to the measured angles, and are explicit that they are modeled structural results rather than precision derivations. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

Scope of this paper

CLAIM: structural estimates of the PMNS mixing angles from rope knot-phase geometry.

NOT CLAIMED: precision derivation of the PMNS matrix. Status: Modeled / partially Open.

1. Mixing as Relative Knot Geometry

In the rope picture, mixing between neutrino states reflects the relative orientation of their knot phases. The mixing angles are read from this geometry, giving structural estimates rather than free parameters.

2. Status

The estimates are of the right general magnitude for the measured mixing angles but are not claimed to precision. They are reported as modeled structural results, with the quantitative gaps to the measured PMNS values left open.

Status

PMNS angle structural estimates — Modeled.

Precision PMNS derivation — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.